

Robert Hayes

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Summary

A dynamic and driven geoscientist with significant experience in programming and IT. Extensive experience in both field data collection and data analysis, using primarily Python to extract insights from data. Confident independent field worker with more than 8 months solo fieldwork experience, primarily in sedimentary logging from PhD. Highly proficient in Python, having built applications for data visualisation and data management, typically with the goal of automating previously manual processes. Additionally proficient in the use of GIS products (ArcGIS, QGIS) and highly adaptable to new technologies, having utilised both numerical modelling and AI within projects. Interested in applying fieldwork and reporting skills to geological engineering, ground investigation or geotechnical questions.

Experience

British Ice Skating, Technical Operations Officer

Sheffield, UK (Remote)

June 2024 – present

- Develop desktop applications using Python to streamline competition management, using libraries including PyQt6, FastAPI, numpy and Pandas/Polars.
- Manage and maintain a large variety of IT and audio-visual equipment, and deploy said equipment in challenging conditions at events.
- Effectively prioritise administration tasks, including event scheduling and database management.

Friedrich Alexander Universität, Postdoctoral Researcher

Erlangen, Germany

Feb 2023 – Feb 2024

- Reconstructed paleoenvironments from fluvial deposits using data collected from fieldwork and drill cores.
- Designed efficient work flows using ArcGIS Pro and Python to increase sediment flux modelling iterations by a factor of 10.
- Wrote tools to leverage AI image recognition for the analysis of grain size distribution in thin sections.

Department of Geosciences, University of Arizona, Graduate Assistant

Tucson, AZ, USA

Aug 2018 – Jan 2023

- Taught lectures and labs in sedimentology to undergraduate classes, receiving unanimously positive feedback from students.
- Organised field trips for up to 50 students, including ensuring health and safety of students was properly managed, receiving more than 80% positive feedback during student evaluation.
- Balanced multiple parallel projects, including student engagement, examination and academic research projects.

Department of Geosciences, University of Arizona, Machine Learning Research Project Assistant

Tucson, AZ, USA

June 2019 – Aug 2019

- Contributed to training of machine learning model investigating links between volcanism and climate change.

University of Leicester AccessAbility Centre, Note Taker

Leicester, UK

Oct 2017 – May 2018

- Prepared comprehensive yet concise notes on a variety of subjects to assist students with additional learning access requirements.
- Developed ability to synthesise important parts of unfamiliar topics when taking notes.

Department of Geosciences, University of Arizona, Paleoclimate Lab Assistant

Tucson, AZ, USA
Jan 2017 – May 2017

- Processed 600 samples for Loss On Ignition analysis, managing and organising sample records.
- Quickly learned and adhered to health and safety regulations and laboratory best practices.

Education

PhD University of Arizona, Geoscience

Tucson, AZ, USA
Aug 2018 – Jan 2023

- Graduated with GPA of 3.9.
- Developed specialist knowledge of fluvial sedimentology through extensive fieldwork and literature research.
- Measured more than 6 km of sedimentary logs at fine scale detail and collected more than 5,000 paleocurrent measurements to build paleogeographic reconstructions of the southwestern US.
- Earned a total of \$27,000 grants and scholarships from Geological Society of America, Arizona Geological Society and University of Arizona.

MGeol. University of Leicester, Geology with Palaeobiology

Leicester, UK
Sept 2014 – July 2018

- Graduated with First Class Honours.
- Analysed behaviour of plastic in geological environment, using pyrolysis GCMS and infinite focus microscopy.
- Performed principal component analysis and other parametric statistical tests on complex textural datasets using R and SPSS.
- Won East Midlands Geological Society Beryl Whittaker Award for best dissertation.

Volunteering

- Organised junior education section at Tucson Gem and Mineral Show in 2020 and 2022, interacting with hundreds of guests at the largest gem trade show in the world. Operated successful event with very limited budget.
- Acted as Audio/Visual technician for GeoDaze 2022, a geological conference held at the University of Arizona, ensuring the smooth operation of technology during presentations.
- Served as treasurer for the Association of Women Geoscientists (AWG) Southern Arizona Chapter during 2021 and 2022, managing funds and creating an automated system for reporting budgets to the central AWG organisation.

Skills

IT: Microsoft Windows, Microsoft Office, Linux (RHEL-derivatives), Python, R, basic Rust, basic MATLAB, ESRI ArcGIS Pro, QGIS, GDAL, Adobe, GIMP, Inkscape, Sony Vegas.

Geology: clastic sedimentology, specialising in fluvial systems; 8 months combined solo fieldwork experience in western USA, with further experience in teams in China and US.

General: grant/bid and report writing experience from preparing academic papers and funding applications.

Publications and Talks

- 2 *first author publications*: Hayes et al. 2025, GSA Bull. doi: 10.1130/B37816.1, Hayes and DeCelles, 2022, JSR, doi: 10.2110/jsr.2021.122.
- 1 first author manuscript currently in preparation.
- 1 invited talk at China University of Geosciences, Wuhan, China.
- 6 conference talks and poster presentations, including Geological Society of America and International Conference on Fluvial Sedimentology.

Miscellaneous

- Full manual UK Class B driver's license (car owner).